

ElementalTask-RML: Emergence-Order Monitoring Extensions for ElementalTask

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<https://github.com/thinkthoughts/ElementalTask-RML>

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Abstract

ElementalTask-RML extends the ElementalTask framework with monitoring-oriented analysis workflows for emergence-order tracking, function-vector drift monitoring, forecast comparison, and cross-model stability evaluation.

Rather than treating emergence as a sudden or opaque phenomenon, this work approaches emergence as a measurable ordering process across constrained evaluation trajectories. The framework combines lightweight monitoring heuristics with RML-inspired monitoring abstractions and CGCS-oriented stability measurements to produce interpretable emergence diagnostics suitable for iterative experimentation and checkpoint-oriented analysis workflows.

Initial experiments demonstrate interpretable emergence-order trajectories, checkpoint drift signatures, forecast comparisons, and cross-model rank-correlation structures which may support future emergence-monitoring workflows built on top of the ElementalTask evaluation setting.

1 Introduction

ElementalTask introduces a framework for studying emergent task behavior across checkpoint progression and task families, motivated by the Implicit Curriculum Hypothesis [1].

ElementalTask-RML is a monitoring-oriented fork and extension of the ElementalTask framework, adding emergence-order tracking, function-vector drift monitoring, forecast comparison, and cross-model stability analysis workflows.

1.1 Upstream Framework

ElementalTask-RML is a monitoring-oriented fork and extension of the ElementalTask framework [1]. The repository adds emergence-order tracking, function-vector drift monitoring, forecast comparison, and cross-model stability analysis workflows on top of the original evaluation structure.

Upstream ElementalTask paper:

<https://arxiv.org/abs/2604.08510>

Upstream ElementalTask repository:

<https://github.com/KaiserWhoLearns/ElementalTask>

Large language model evaluation increasingly involves longitudinal checkpoint analysis, evolving task distributions, and monitoring workflows. Emergent task behavior has consequently become an important topic in language model evaluation and scaling studies [2].

This repository investigates whether emergence ordering exhibits measurable structural consistency across checkpoint progression.

Ordering behavior is operationally useful because instability, recurrence, and convergence patterns may appear before stable benchmark saturation.

The repository focuses on lightweight monitoring tools capable of:

- tracking emergence ordering across checkpoints,
- measuring function-vector drift trajectories,
- comparing cross-model emergence alignment,
- monitoring checkpoint stability signals,
- visualizing constraint-oriented task structure.

The goal is not to replace benchmark evaluation, but to *augment* ElementalTask with lightweight monitoring and ordering diagnostics across checkpoint progression.

2 Monitoring Framework

2.1 Emergence-Order Tracking

The monitoring pipeline evaluates task-behavior appearance ordering across ElementalTask checkpoints and task families. Rather than focusing exclusively on final benchmark values, the framework emphasizes relative ordering structure and trajectory consistency.

Emergence ordering is treated as a dynamic process:

observe $\rightarrow$ interpret $\rightarrow$ monitor $\rightarrow$ forecast $\rightarrow$ compare

This ordering perspective enables monitoring systems to analyze:

- stability of task-behavior rank transitions,
- recurrence of emergent patterns,
- checkpoint consistency structure,
- divergence across model families.

2.2 Drift Monitoring

Function-vector trajectory monitoring measures changes across checkpoints and evaluation stages.

Drift analysis is useful for identifying:

- unstable transition regions,
- sudden ordering discontinuities,
- convergence structure,
- divergence signatures.

The framework emphasizes interpretable trajectory monitoring rather than black-box aggregate scoring alone.

2.3 Constraint-Oriented Stability

The repository incorporates CGCS-inspired monitoring heuristics emphasizing checkpoint stability and constraint-oriented coherence.

Repeated emergence patterns are treated as stronger indicators of stable emergence than isolated high-variance observations.

3 Monitoring Results

3.1 Emergence-Order Monitoring

Figure 1 illustrates emergence-order trajectories across monitored ElementalTask task families.

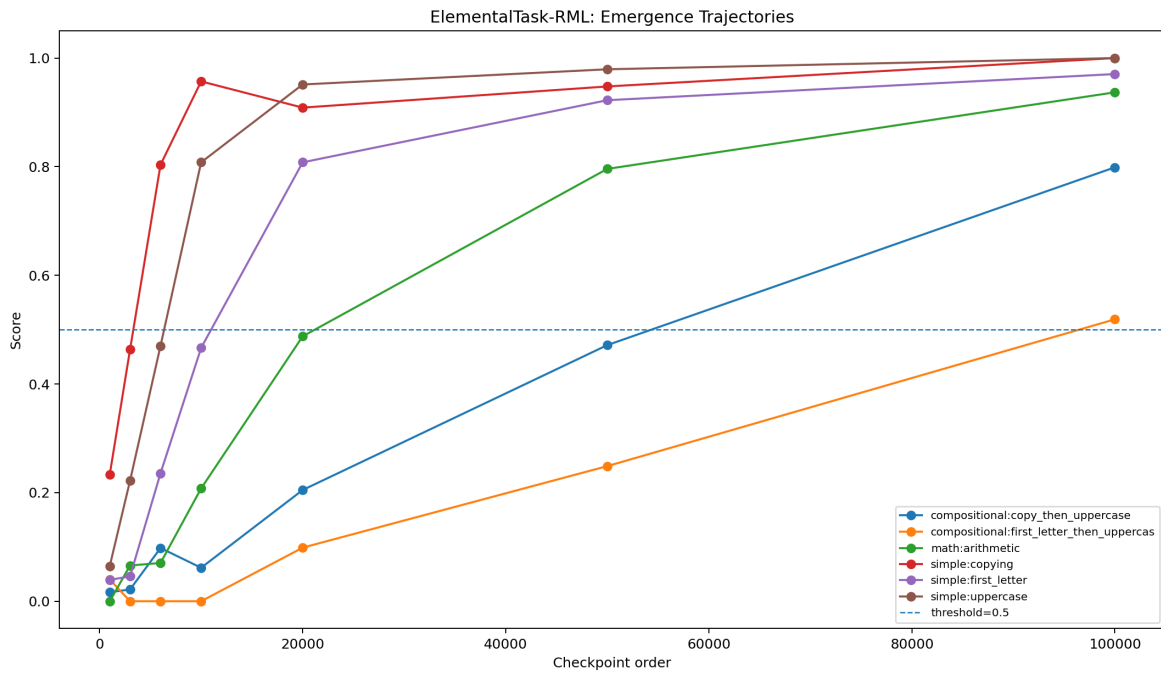


Figure 1: Emergence-order monitoring across ElementalTask task families and checkpoints.

3.2 Function-Vector Drift

Figure 2 visualizes function-vector drift trajectories during checkpoint progression.

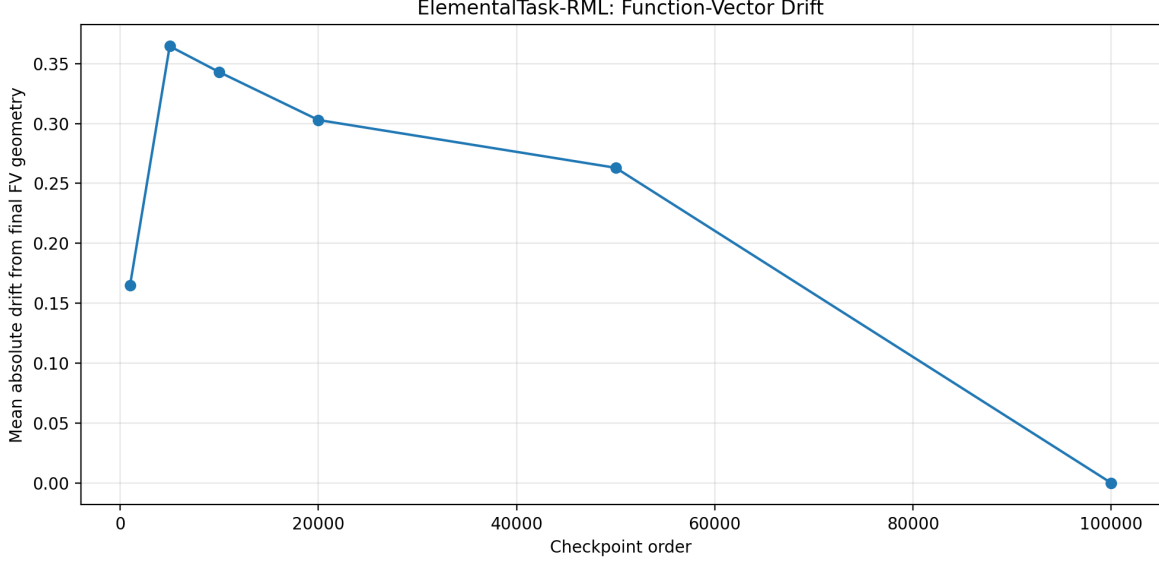


Figure 2: Function-vector drift monitoring across checkpoints.

3.3 Constraint Drift Flags

Figure 3 summarizes constraint drift flag behavior and monitoring thresholds.

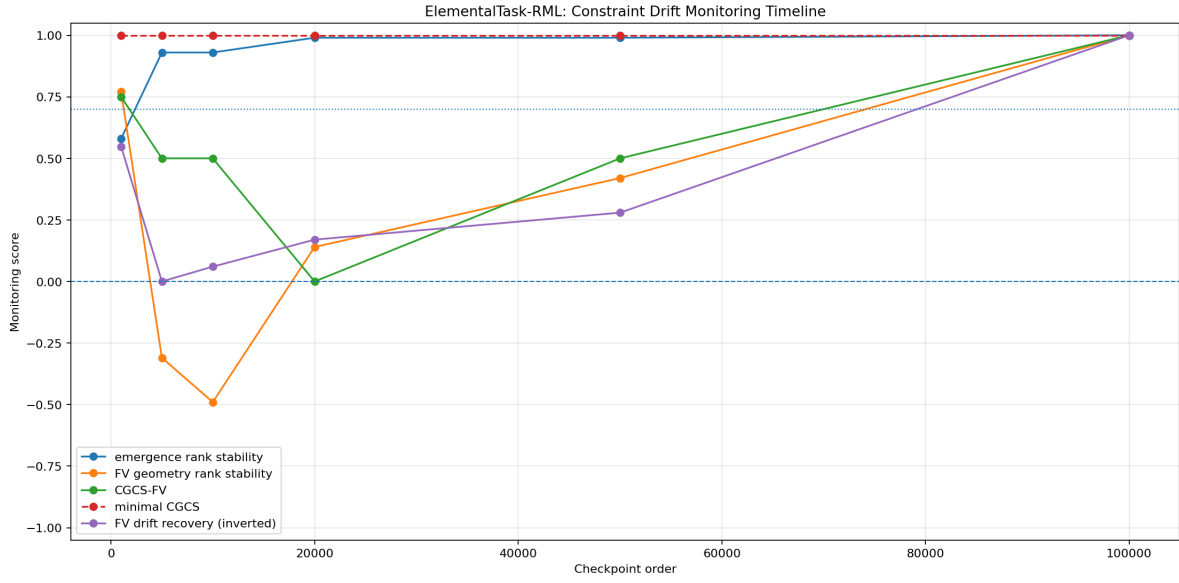


Figure 3: Constraint drift flags and monitoring thresholds.

3.4 Forecast vs Actual

Figure 4 compares simple emergence forecasts against observed checkpoint trajectories.

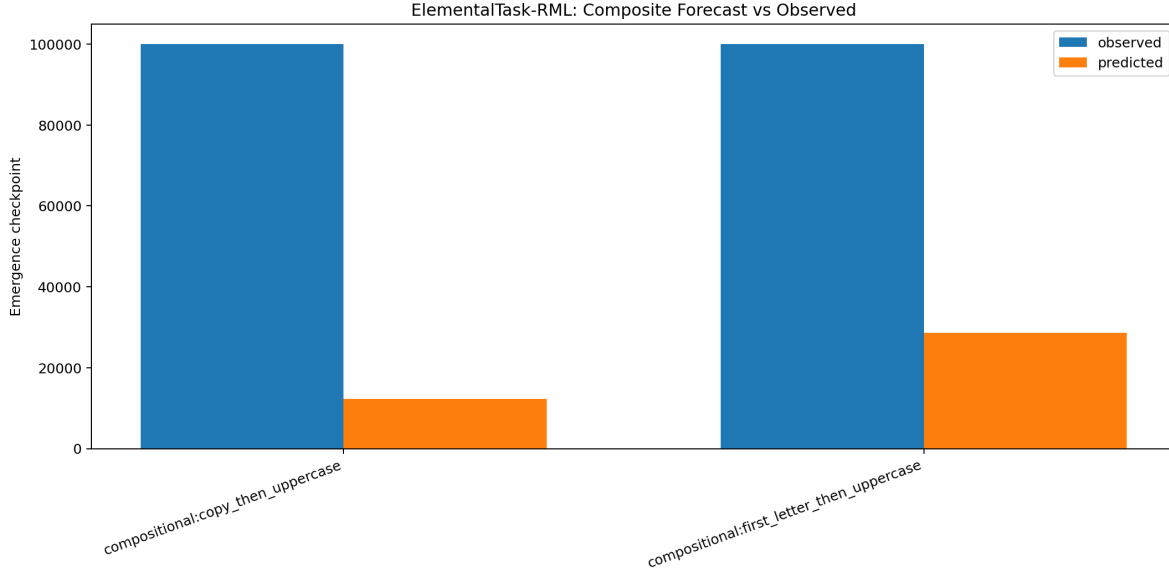


Figure 4: Composite emergence forecast versus observed checkpoint behavior.

3.5 Cross-Model Rank Correlation

Figure 5 visualizes cross-model emergence rank correlation structure.

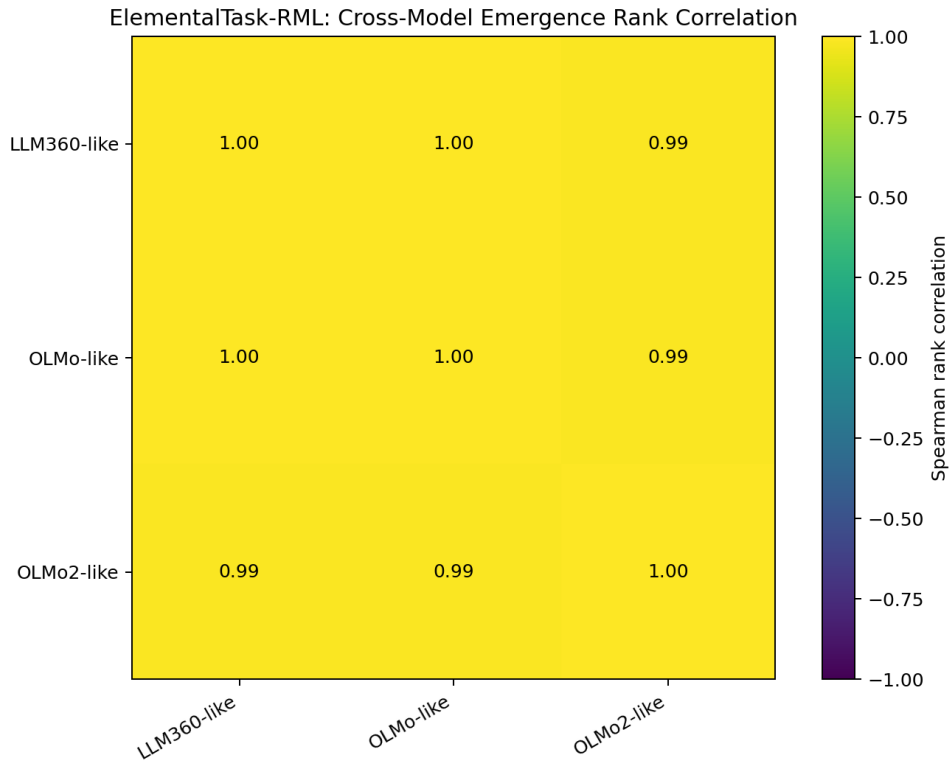


Figure 5: Cross-model emergence rank correlation structure.

3.6 Multi-Model Stability

Figure 6 monitors longitudinal emergence-order stability across checkpoints.

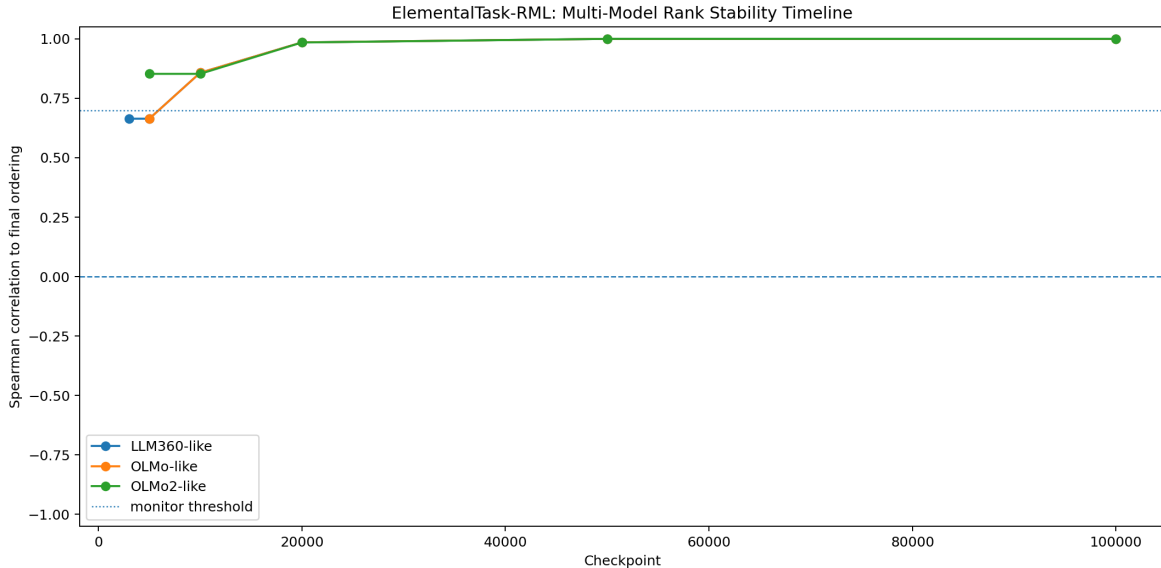


Figure 6: Multi-model emergence-order stability across checkpoint progression.

4 Interpretation

Emergence is treated here less as a sudden phase transition and more as an interpretable monitoring problem involving recurrence, ordering, and structural persistence.

Within the ElementalTask evaluation setting, these monitoring outputs suggest that emergence ordering exhibits measurable structural consistency across checkpoints and model families.

Several monitoring patterns motivate continued investigation:

- emergence trajectories frequently exhibit measurable structure,
- checkpoint stability may precede reliable benchmark convergence,
- drift signatures often reveal transition instability,
- cross-model comparisons expose alignment and divergence regions.

The resulting monitoring perspective emphasizes emergence as a longitudinal measurement problem rather than a singular benchmark event.

5 Conclusion

ElementalTask-RML extends ElementalTask with monitoring-oriented analysis workflows emphasizing:

- emergence-order tracking,

- checkpoint stability,
- checkpoint drift analysis,
- cross-model comparison,
- interpretable structural visualization.

The resulting framework provides a lightweight foundation for future experiments involving checkpoint-oriented analysis and emergence-order diagnostics built on top of the ElementalTask evaluation ecosystem.

Emergence is therefore framed less as a singular benchmark event and more as a longitudinal measurement problem involving recurrence, ordering, and stability over checkpoint progression.

Repositories

ElementalTask-RML repository:

<https://github.com/thinkthoughts/ElementalTask-RML>

ElementalTask upstream repository:

<https://github.com/KaiserWhoLearns/ElementalTask>

References

- [1] Emmy Liu, Kaiser Sun, Millicent Li, Isabelle Lee, Lindia Tjumatja, Jen-tse Huang, and Graham Neubig. What do language models learn and when? the implicit curriculum hypothesis. *arXiv preprint arXiv:2604.08510*, 2026.
- [2] Jason Wei et al. Emergent abilities of large language models. *Transactions on Machine Learning Research*, 2022.